

MINJI SON

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RESEARCH INTERESTS

Resource management and performance predictability for ML workloads on shared GPUs, with a focus on interference-aware scheduling across GPU sharing mechanisms (MPS, MIG, time-slicing) and tail latency SLOs for LLM inference serving

EDUCATION

Sungkyunkwan University, South Korea Mar 2015 – Aug 2019
B.S. in Software Engineering, Department of Computer Science and Engineering

- GPA: 3.72 / 4.5 (92.2 / 100); Major GPA: 3.93 / 4.5
- Operating Systems, Computer Architectures, System Programming, Compiler Design

RESEARCH EXPERIENCE

Undergraduate Research Assistant, ARCS Lab, Sungkyunkwan University Mar 2017 – Dec 2018
Advisor: Prof. HwanSoo Han

GPU Performance in Containers and Multi-Task Scheduling (First author, KSC 2018)

- Measured execution time, memory usage, and multi-task scheduling performance of NVIDIA Docker 2.0.2 using Rodinia 3.1 on a GTX 980 Ti
- Identified a constant 0.96 s overhead independent of the application, attributing the cost to the NVIDIA Docker software stack rather than GPU execution
- Determined the device memory limit under GPU container co-location by scaling concurrent containers, with failures beginning at approximately 93% memory utilization
- Compared execution time with and without NVIDIA MPS for co-scheduled kernels, classifying cache sensitivity based on the change in L1 hit rate with doubled effective cache capacity
- Found that MPS degraded performance by up to 3.6x when cache sensitive kernels saturated the cache, and proposed checking cache utilization before enabling MPS

Spark Performance Analysis and Task Execution (KSC 2017)

- Measured Spark 2.1.0 job time on a 3-node cluster with 3 executor configurations and 4 input sizes to examine how executor memory should be sized relative to input size
- Found that configurations with more cores ran faster at 50GB and 75GB but slower at 100GB where the input exceeded memory available per node
- Traced the 100GB inversion to JVM garbage collection since each task processes a fixed 128MB block regardless of executor memory and 2GB executors spent more time in GC

PUBLICATIONS

[1] **Minji Son**, Hyunjun Kim, and Hwansoo Han, "Analyzing GPU Performance on Containers and Characteristics of Multi-Task Scheduling," Korea Software Congress (KSC), 2018 (in Korean).

[2] Minseop Jeong, Dongeun Lee, **Minji Son**, Jeonghwan Park, and Hwansoo Han, "Performance Analysis of Spark Application through Task Execution," Korea Software Congress (KSC), 2017 (in Korean).

WORK EXPERIENCE

Samsung Electronics, Mobile eXperience Division — Software Engineer Jul 2021 – Present
One of three HQ engineers developing core components of the Samsung Messaging Platform, a push notification platform serving 1.26B Galaxy devices and 600M+ pushes per day. Led technical design for a team of seven engineers in Vietnam

Distributed query optimization (Amazon Redshift)

- Optimized Redshift queries joining 6 tables of 1.5-3.3 billion rows each after they slowed to 6-10 min

- Explained an 8x runtime gap between two versions of one query by tracing it to a single extra column in a DISTINCT subquery, which doubled the intermediate row count
- Found that the optimizer misestimated row counts on the largest table by 31x, then reordered the joins to filter that table before joining it, removing cross-node broadcasts
- Brought every slow case under 1 min (**7-13x faster**) and confirmed identical results for each, without slowing cases that were already fast
- Reduced a per-service statistics batch on Redshift from **~2.6 h to ~1.2 h** by precomputing an expensive shared join as a materialized view, and applied it only to large services after measuring that the view's overhead slowed small ones

Multimodal AI integration and infrastructure automation

- Integrated a multimodal AI moderation service using an LLM, OCR, and object detection into the content approval pipeline.
- Designed asynchronous processing to handle variable AI response times and added human review for failed or uncertain results, automating 87% of content reviews
- Built a Terraform-based service to provision AWS/Azure infrastructure with internal security policies applied by default, reducing provisioning time from days to under 1 hour

Korea Telecom, Connected Car Platform Development Team — Software Engineer Jul 2019 – Jul 2021

- One of 2 engineers owning the third-party integration layer of a global automotive OEM's membership service; developed and operated 40+ Java/Spring APIs connecting to 6 external providers

Samsung Electronics, Device Solutions — Intern Jul 2018 – Aug 2018

- Improved SSD failure analysis tools in C++ automating the dump loading workflow and reducing manual steps from 7 to 1
- Automated diagnostic summarization to reduce the manual effort required to inspect SSD failure dumps

PATENTS

Co-inventor of 7 Samsung Electronics patent applications (KR/US/EP, 2023) in on-device ML applications including handwriting recognition and content filtering

TEACHING EXPERIENCE

Teaching Assistant, Compiler Design (Prof. Hwansoo Han), Sungkyunkwan University Spring 2018

TA, Computational Thinking and SW Coding, Sungkyunkwan University Spring 2017

PROFESSIONAL ACTIVITIES

WITS (Women in Technology at Samsung), Samsung Electronics Jan 2025 – Dec 2025

- Participated in a company wide mentoring program with senior women leaders in engineering

WISSET Engineering Research Program, Korea Center for Women in Science Apr 2017 – Oct 2017

- Select for a government-funded research program supporting 100 teams nationwide
- Investigated zswap, a Linux compressed swap mechanism, in mobile environments and explored ways to reduce response time

SKILLS

- **Systems & Infrastructure:** Linux, Docker, Kubernetes, Terraform, CUDA, Apache Spark
- **Cloud (AWS):** EKS, Redshift, EC2, Lambda, CloudWatch
- **Programming & Backend:** Java, C, C++, Python; Spring Boot, Spring Batch, MySQL, PostgreSQL
- **Languages:** Korean (native), English (fluent, TOEFL 110), Japanese (conversational)